Unbounded Failure of Horizonwise Optimality for Exact Causal Seeds

Jan Mikulík

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Abstract

We construct finite controlled trees of depth two for which both horizon optima equal their respective information-envelope lower bounds, but no globally optimal realization restricts to an optimal realization of the first horizon. With m root actions and binary root outputs, the first-horizon optimum is one bit, whereas every globally optimal seed induces a first-response table of entropy at least $m - \log_2(3/2)$ bits. The discrepancy is therefore unbounded. The obstruction arises from distinct atom masses: equality in entropy data processing forces all full-policy transcripts to identify the same latent state. The theorem concerns exact minimizers; historical priority and Lean formalization are not asserted.

1 Setting and the relevant restriction

A finite controlled tree prescribes, at each nonterminal history, a finite collection of actions and a probability law for the output of each action.

An exact causal realization consists of a discrete seed S and deterministic responses depending on the seed, the observed history, and the chosen action. Under every deterministic adaptive policy, the resulting transcript must have its prescribed law.

Let C_n denote the minimum seed Shannon entropy for exact simulation through horizon n. All entropies and logarithms are in base two.

Suppose the root has actions $1, \dots, m$. For a realization with seed S, write

$$A_i(S)$$

for its root response to action i, and define its first-response table by

$$R_1(S) = (A_1(S), \dots, A_m(S)).$$

The entropy $H(R_1(S))$ measures the law of the strategy restricted to the first horizon. It does not count seed information unused by all first-horizon responses.

The distinction is essential: the theorem below concerns the entropy of this actual restricted strategy, not merely the entropy of an unnecessarily retained seed.

For completeness, let L_n denote the mean of the lower envelope of the policy transcript-information CDFs:

$$F_n^*(t) = \min_{\pi} \mathbb{P}[-\log P_{n,\pi}(Y) \le t], \qquad L_n = \int_0^\infty (1 - F_n^*(t))\, dt.$$

2 Main theorem

Theorem 1 (Unbounded restriction gap)**.** *For every integer $m \geq 3$, there exists a controlled tree T_m of depth two satisfying the following properties:*

1. *The root has m actions, each with a fair binary output. After every root action and output, there is a single continuation action.*
2. *Every prescribed probability on every local output alphabet is strictly positive.*
3. *Both horizon optima attain their information-envelope bounds:*

$$C_1 = L_1 = 1, \qquad C_2 = L_2.$$

4. *Every exact realization S attaining $H(S) = C_2$ satisfies*

$$\boxed{H(R_1(S)) = C_2 \geq m - \log_2(3/2).}$$

Consequently,

$$H(R_1(S)) - C_1 \geq m - 1 - \log_2(3/2) > m - 2.$$

In particular, no realization is optimal both globally and after restriction to the first horizon.

3 A balanced law with distinct atom masses

Let

$$V_m = \{-1, 1\}^m.$$

Define characters

$$\chi_j(z) = z_1 z_{j+2} \quad (0 \leq j \leq m-2), \qquad \chi_{m-1}(z) = z_1 z_2 z_3,$$

and put

$$M_m = \sum_{j=0}^{m-1} 3^j = \frac{3^m - 1}{2}.$$

For $z \in V_m$, define

$$p_m(z) = 2^{-m} \left(1 + \frac{1}{2M_m} \sum_{j=0}^{m-1} 3^j \chi_j(z) \right). \tag{1}$$

Lemma 1. *The function p_m is a probability law satisfying:*

1. *every coordinate is fair;*
2. *all its atom masses are positive and pairwise distinct;*
3. $$H(p_m) \geq m - \log_2(3/2) > m - 1.$$

Proof. Under the uniform law on V_m, every nonconstant sign character has mean zero. Each χ_j has degree at least two, so it is orthogonal both to the constant character and to every coordinate character z_i. Summing (1) therefore gives

$$\sum_z p_m(z) = 1, \qquad \sum_z z_i p_m(z) = 0 \quad (1 \leq i \leq m).$$

The second identity is equivalent to fairness of coordinate i.

Since $|\chi_j(z)| = 1$,

$$2^{-m-1} \le p_m(z) \le \frac{3}{2}\, 2^{-m}.$$

In particular, all masses are positive and

$$-\log p_m(z) \ge m - \log_2(3/2).$$

Taking expectation proves the entropy bound.

It remains to prove distinctness. The pair characters give

$$a_i = z_1 z_i, \qquad 2 \le i \le m,$$

and the triple character gives $b = z_1 z_2 z_3$. They recover the vector through

$$z_1 = b a_2 a_3, \qquad z_i = z_1 a_i.$$

Thus the character vector distinguishes all points of V_m.

Furthermore, the map

$$(c_0, \dots, c_{m-1}) \longmapsto \sum_{j=0}^{m-1} 3^j c_j, \qquad c_j \in \{-1, 1\},$$

is injective. Indeed, if two sign vectors differ and r is their largest differing index, the contribution of that index to the difference has magnitude $2 \cdot 3^r$, whereas all smaller indices contribute at most

$$2 \sum_{j<r} 3^j = 3^r - 1.$$

Cancellation is impossible. Equation (1) consequently assigns distinct masses to distinct points. □

4 Construction of the controlled tree

Fix $m \ge 3$ and abbreviate $p = p_m$.

At the root, the available actions are $i \in \{1, \dots, m\}$. Action i returns a fair sign $b \in \{-1, 1\}$.

After history (i, b), there is one continuation action. Its output alphabet is

$$V_{i,b} = \{z \in V_m : z_i = b\},$$

and its kernel is

$$K_{i,b}(z) = 2p(z), \qquad z \in V_{i,b}. \tag{2}$$

Coordinate fairness implies

$$\sum_{z \in V_{i,b}} K_{i,b}(z) = 1.$$

Every entry is positive.

Under the policy choosing root action i, the complete transcript (b, z) has $b = z_i$ and probability

$$\frac{1}{2}\, 2p(z) = p(z).$$

Thus each complete policy transcript is equivalent to a sample from the same law p.

Lemma 2. *For this tree,*

$$C_1 = L_1 = 1, \qquad C_2 = L_2 = H(p).$$

Proof. Every root action produces a fair sign, so every exact first-horizon seed has entropy at least one. One common fair sign, returned for every root action, attains this bound. Every first-horizon information variable is identically one, hence $L_1 = 1$.

At horizon two, every policy transcript has entropy $H(p)$. Every exact seed must therefore have entropy at least $H(p)$.

To attain this value, take a seed $Z \sim p$. On root action i, return Z_i. On the resulting history (i, Z_i), return Z. To make the strategy total, define the continuation response at each incompatible history (i, b), $b \neq Z_i$, to be any fixed member of $V_{i,b}$. Such histories are not reached by that seed value.

This realization is exact and has entropy $H(p)$. All policy transcript-information laws are identical, so their lower envelope has mean $H(p)$ as well. □

5 Rigidity of every global optimum

Lemma 3 (Equality rigidity for distinct masses)**.** *Let p be a strictly positive law on a finite set V, with pairwise distinct atom masses. Suppose S is discrete and $Z^{(1)}, \dots, Z^{(m)}$ are deterministic functions of S satisfying*

$$\mathcal{L}(Z^{(i)}) = p, \qquad H(S) = H(p).$$

Then

$$Z^{(1)} = \cdots = Z^{(m)} \quad \textit{almost surely.}$$

Proof. For each i, deterministic decoding and the entropy chain rule give

$$H(S) = H(Z^{(i)}) + H(S \mid Z^{(i)}).$$

Thus $H(S \mid Z^{(i)}) = 0$.

For every $z \in V$, conditional on $Z^{(i)} = z$, the seed is almost surely a single value. Since $p(z) > 0$, this means that exactly one positive-mass seed atom maps to z, and its mass is $p(z)$.

Now fix a positive-mass seed value s. If $Z^{(i)}(s) = z$ and $Z^{(j)}(s) = z'$, then

$$p(z) = \mathbb{P}(S = s) = p(z').$$

Distinctness of the atom masses forces $z = z'$. There are only finitely many indices i, j, so the asserted simultaneous equality holds almost surely. □

Proof of Theorem 1. The structural and positivity assertions follow from (2), and the optimum identities follow from Lemma 2.

Let S be any exact realization with $H(S) = C_2 = H(p)$. For root action i, let $Z^{(i)}$ be its second-step output. Exactness gives $\mathcal{L}(Z^{(i)}) = p$.

By Lemma 3,

$$Z^{(1)} = \cdots = Z^{(m)} =: Z \quad \text{almost surely.}$$

The continuation output belongs to $V_{i,A_i(S)}$, so

$$A_i(S) = Z_i^{(i)} = Z_i.$$

Therefore

$$R_1(S) = Z \quad \text{almost surely,}$$

and hence

$$H(R_1(S)) = H(p) = C_2.$$

Lemma 1 proves the quantitative bound. □

6 Consequences

Corollary 1 (No uniform bound for optimal restrictions)**.** *There is no finite constant K such that every finite controlled tree admits a globally entropy-optimal realization whose restriction to the first horizon has entropy at most $C_1 + K$.*

Proof. For T_m, every globally optimal realization has restriction entropy at least $m - \log_2(3/2)$, while $C_1 = 1$. Let $m \to \infty$. □

Corollary 2 (Zero envelope overhead does not imply compatible minimizers)**.** *The identities $C_1 = L_1$ and $C_2 = L_2$ do not imply the existence of an exact second-horizon minimizer whose first-horizon restriction is a first-horizon minimizer.*

Proof. Apply Theorem 1. □

One may extend T_m indefinitely by singleton deterministic outputs after its second step. Then

$$C_\infty = \sup_n C_n = C_2,$$

yet every infinite-horizon minimizer still has the same nonoptimal first-response table. This is fully consistent with the projective entropy identity: that identity concerns optimal values, not simultaneous optimality of the restrictions of an optimizer.

7 Scope and limitations

The result uses m root actions and 2^{m-1} possible outputs at each continuation node. It does not establish an unbounded gap with a uniformly bounded number of actions and uniformly bounded output alphabets.

The rigidity argument concerns exact entropy minimizers. It does not provide a uniform quantitative obstruction for seeds allowed a fixed additive excess above C_2. The distinct atom masses become finely separated as m increases, so such a robustness conclusion requires a separate argument.

Although the construction is a controlled tree with a shared causal decoder, its depth-two form also admits a description as a minimum-entropy coupling problem followed by deterministic coarsenings. The elementary ingredients—entropy data processing, equality rigidity, and balanced sign characters— are not claimed as new.

The precise structural conclusion is the unbounded failure of horizonwise optimality despite equality with the information envelope at both horizons. Its historical priority has not been established, and the proof has not been formalized in Lean.

References

[1] J. Mikulík, *One Seed for Every Adaptive Policy.* Mathematical notes and Lean development. `https://github.com/jendamikulik/finite-Shannon-theorem`.